STPM 2022 PHYSICS 960/3

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, 1 u $= 1.66 \times 10^{-27}$ kg.

Section A [15 marks]

1. An object of mass 0.10 kg undergoes simple harmonic motion along the x-axis according to the equation $\frac{d^2x}{dt^2} + 100x = 0$. If the total energy of the system is 0.0020 J, what is the amplitude of the motion?

A 0.040 cm B 0.20 cm C 0.40 cm D 2.0 cm

Answer: D

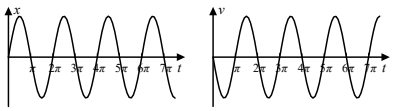
$$\omega^2 = 100 \Longrightarrow \omega = 10 \text{ rad s}^{-1}$$

$$E = \frac{1}{2}m\omega^2A^2 \Longrightarrow 0.0020 = \frac{1}{2}(0.10)(100)A^2 \Longrightarrow A = 0.020 \text{ m} = 2.0 \text{ cm}$$

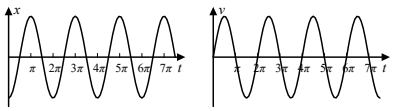
Explanation: Compare with $d^2x/dt^2 = -\omega^2x$. The total energy equals the maximum kinetic energy.

2. Which graph of displacement, x, against time, t, corresponds correctly to the graph of velocity, v, against time, t, for a simple harmonic motion?

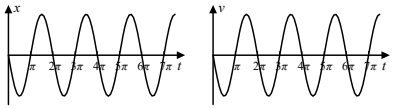
A



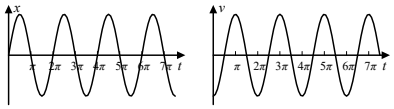
B



C



D



Answer: B

If $x = A \sin \omega t$, then $v = A\omega \cos \omega t$: the velocity is a quarter-cycle ahead, at its maximum when $x = 0$ and increasing. This matches B.

Explanation: The velocity is the gradient of the x–t graph: maximum where x crosses zero going upwards, zero at the peaks of x.

3. The principle of superposition states that when two or more waves overlap at one point at the same instant, the resultant displacement is equal to

A the sum of the individual amplitude of the waves
B the sum of the individual displacement of the waves
C the product of the individual amplitude of the waves
D the product of the individual displacement of the waves

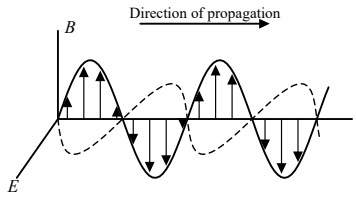
Answer: B

The resultant displacement is the (vector) sum of the individual displacements of the waves at that point.

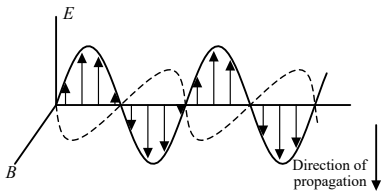
Explanation: Amplitudes add only where the waves are in phase; the principle is about instantaneous displacements.

4. Which graph shows the correct direction of propagation of an electromagnetic waves?

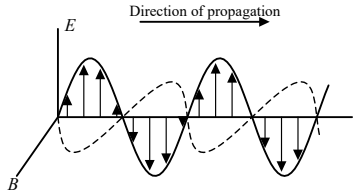
A



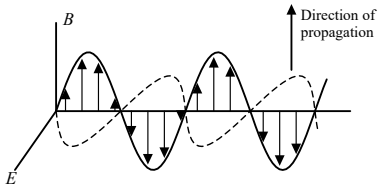
B



C



D



Answer: C

An electromagnetic wave travels in the direction of **E** × **B**: with **E** vertical and **B** along the axis towards the viewer, the wave travels along the horizontal axis as in C. **E** and **B** are both perpendicular to the direction of propagation.

Explanation: In a transverse electromagnetic wave the direction of travel cannot be parallel to either field (B and D are wrong).

5. The sound intensity level of 88.0 dB is detected 30.0 cm from a point source. What is the amount of energy received at the point in one second?

[The threshold intensity of hearing is 1.00×10^{-12} W m^{-2}.]

A 11.9×10^{-4} J B 7.14×10^{-4} J C 5.58×10^{-4} J D 1.78×10^{-4} J

Answer: B

$$I = (1.00 \times 10^{-12})10^{8.80} = 6.31 \times 10^{-4} \text{ W m}^{-2}$$

$$P = I(4\pi r^2) = (6.31 \times 10^{-4})(4\pi)(0.300)^2 = 7.14 \times 10^{-4} \text{ W} \Longrightarrow 7.14 \times 10^{-4} \text{ J in 1 s}$$

Explanation: For a point source the energy spreads over a sphere of area $4\pi r^2$; the answer is the energy per second crossing that sphere.

6. A musician plucks simultaneously the first two strings of a ukulele and hears three beats per second. The first string is tuned to 350 Hz. As the musician tightens the screw to stretch the second string and plucks the two strings again, the musician hears two beats per second. What is the frequency of the second string before it is stretched?

A 345 Hz B 347 Hz C 351 Hz D 353 Hz

Answer: B

Three beats: the second string is 347 Hz or 353 Hz.

Tightening raises its frequency. The beats fall to 2 per second, so it moved closer to 350 Hz: it was 347 Hz (now 348 Hz).

Explanation: If it had been 353 Hz, tightening would increase the beat frequency.

7. An object is placed 7.0 cm in front of a concave mirror of focal length 15 cm. The object is then moved to the position of 21 cm in front of the mirror. Which is true about the characteristics of the first and second images formed?

A From a virtual image to a real image
B From a real image to a virtual image
C From a small image to a large image
D From an inverted image to an upright image

Answer: A

At 7.0 cm (inside $f = 15$ cm) the image is virtual and upright; at 21 cm (between F and C) it is real, inverted and magnified.

Explanation: Both images are magnified, so C is wrong; the change is from upright to inverted, so D is wrong.

8. A plano concave lens has a radius of curvature 33.4 cm and refractive index 1.58. What is the focal length of the lens?

A 6.47 cm B 12.9 cm C 28.8 cm D 57.6 cm

Answer: D

$$\frac{1}{f} = (1.58 - 1)\left(\frac{1}{\infty} - \frac{1}{33.4}\right) \Rightarrow f = -57.6 \text{ cm}$$

Magnitude of the focal length: 57.6 cm (diverging lens).

Explanation: One surface is flat (infinite radius). The negative sign shows that a plano-concave lens is diverging.

9. Which statement is **not** true about the Huygens's principle?

A The wavelets are plane waves.
B The principle explains about refraction.
C The principle explains about the law of reflection.
D Every point on a wave front is the source of secondary wavelets.

Answer: A

Huygens's secondary wavelets are spherical, not plane. Statement A is not true.

Explanation: The principle explains reflection and refraction, and every point on a wavefront is a source of wavelets.

10. Which is the reason for the replacement of coaxial cables with optical fibre in modern telecommunications?

A Easily connected
B Chemically stable
C Has low signal loss
D Has high mechanical strength

Answer: C

Optical fibres have very low signal loss (low attenuation), so signals travel long distances with fewer repeaters; they also have a large bandwidth and no electromagnetic interference.

Explanation: Low signal loss is the main technical advantage over coaxial copper cable.

11. When light with photon energy, E, falls on a metal surface, electrons are ejected from the metal surface. Which statement is true about the ejected electrons?

A Electrons are ejected after a lapse of time.
B More electrons are ejected if the light with photon energy is greater than E.
C The kinetic energy of the ejected electron is always less than the photon energy.
D The kinetic energy of the ejected electron increases if the intensity of the light is increased.

Answer: C

$K_{max} = E - W$, so the kinetic energy is always less than the photon energy

Explanation: Emission is instantaneous (A wrong); the number of electrons depends on intensity, not photon energy (B wrong); intensity does not change the kinetic energy (D wrong).

12. An electromagnetic wave of wavelength 220 nm is incident on a gold plate. What is the maximum kinetic energy of the emitted photoelectrons if the gold has a work function of 5.10 eV?

A 0.550 eV B 5.10 eV C 5.65 eV D 10.8 eV

Answer: A

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{220 \times 10^{-9}} = 9.04 \times 10^{-19}\ \text{J} = 5.65\ \text{eV}$$

$$K_{max} = 5.65 - 5.10 = 0.55\ \text{eV}$$

Explanation: Convert the photon energy to eV (divide by 1.60×10^{-19}) before subtracting the work function.

13. What is the kinetic energy of an electron that has a de Broglie wavelength of 2.12×10^{-10} m?

A 3.40 eV B 13.6 eV C 27.2 eV D 33.5 eV

Answer: D

$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}}{2.12 \times 10^{-10}} = 3.13 \times 10^{-24} \text{ kg m s}^{-1}$$

$$K = \frac{p^2}{2m} = 5.37 \times 10^{-18} \text{ J} = 33.5 \text{ eV}$$

Explanation: Use the electron mass and convert joules to electronvolts at the end.

14. The half-life of a radioactive substance is T. The substance consists of N_0 radioactive nuclei at time $t = 0$. What is the time taken for the remaining number of nuclei of the substance to be $\frac{1}{\sqrt{2}}N_0$?

A $\frac{1}{4}T$ B $\frac{1}{\sqrt{2}}T$ C $\frac{1}{2}T$ D $\frac{3}{4}T$

Answer: C

$$N = N_0(\tfrac{1}{2})^{t/T} = \frac{N_0}{\sqrt{2}} = N_0(\tfrac{1}{2})^{1/2} \Longrightarrow t = \frac{1}{2}T$$

Explanation: The fraction $1/\sqrt{2}$ is $(½)^{½}$, so half a half-life has passed.

15. Which is the role of a moderator in a nuclear fission reactor?

A Increases the temperature of the nuclear reactor
B Produces more neutrons during the nuclear reaction
C Absorbs the excess neutrons to produce a chain reaction
D Slows down the neutrons that are produced in the reaction

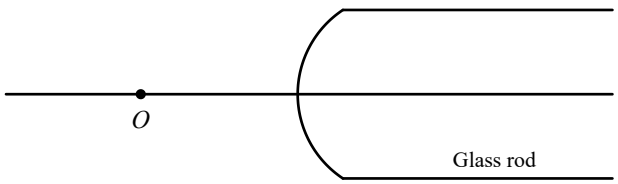
Answer: D

A moderator (e.g. graphite or water) slows down fast neutrons to thermal speeds so they can cause further fission of U-235.

Explanation: Control rods absorb excess neutrons; the moderator only slows them.

Section B [15 marks]

16. An object, O, is placed 6.0 cm in front of a glass rod which has a spherical surface as shown in the diagram below.



The radius of curvature for the glass is 15 cm. The refractive index of the glass is 1.5.

(a) Determine the position of the formed image by refraction. [3]

Answer

$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r}$, with $r = +15$ cm — 1

$\frac{1}{6.0} + \frac{1.5}{v} = \frac{1.5 - 1}{15}$ — 1

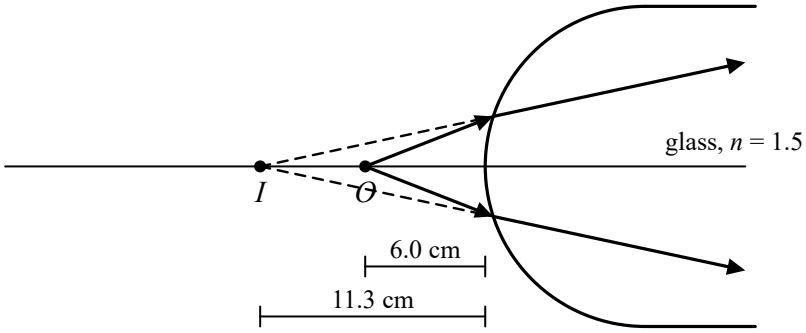
$v = -11.3$ cm: a virtual image 11.3 cm from the surface, on the same side as O — 1

3

Explanation: The object is inside the 'focal length' of the surface, so the rays still diverge after refraction and the image is virtual.

(b) Sketch the ray diagram that shows the formation of the image. [2]

Answer



Rays from O refracted towards the normal at the surface, still diverging in the glass — 1

Refracted rays extended backwards (dashed) to meet at the virtual image I — 1

2

Explanation: A virtual image is where the refracted rays appear to come from.

(c) Determine the position of the formed image by reflection. [3]

Answer

The surface acts as a convex mirror: $f = -\frac{r}{2} = -7.5$ cm — 1

$\frac{1}{v} = \frac{1}{-7.5} - \frac{1}{6.0}$ — 1

$v = -3.33$ cm: a virtual image 3.33 cm behind the surface — 1

3

Total for question **8**

Explanation: Some light is reflected at the curved surface, which bulges towards the object, so it behaves as a convex mirror.

17. An equation for a possible nuclear reaction is given by

$^{4}_{2}\text{He} + {}^{14}_{7}\text{N} \rightarrow {}^{17}_{8}\text{O} + {}^{1}_{1}\text{H}$

Data for the nuclei are given in the table below.

Nuclei	Symbol	Atomic mass/u
Proton	$^{1}_{1}\text{H}$	1.00728
Helium	$^{4}_{2}\text{He}$	4.00260
Nitrogen	$^{14}_{7}\text{N}$	14.00307
Oxygen	$^{17}_{8}\text{O}$	16.99913

(a) The mass of the nucleus of helium is less than the total mass of two protons and two neutrons. Give your reason for the difference. [1]

Answer

When the nucleons combine, some mass (the mass defect) is converted to binding energy that is released — 1

1

Explanation: $E = \Delta mc^2$.

(b) Calculate the mass loss for the reaction. [2]

Answer

Mass of reactants = 4.00260 + 14.00307 = 18.00567 u; products = 16.99913 + 1.00728 = 18.00641 u — 1

$\Delta m = -7.4 \times 10^{-4}$ u $= -1.23 \times 10^{-30}$ kg (mass is gained, not lost) — 1

2

Explanation: The products are heavier than the reactants, so energy must be supplied for the reaction.

(c) Calculate the required energy, in Joule, that enables the reaction to occur. [2]

Answer

$E = \Delta mc^2 = (1.228 \times 10^{-30})(3.00 \times 10^{8})^2$ — 1

$E = 1.11 \times 10^{-13}$ J — 1

2

Explanation: This energy must come from the kinetic energy of the incoming helium nucleus.

(d) Determine the minimum speed of the nucleus of helium that enables the reaction to occur. [2]

Answer

$\frac{1}{2}mv^2 = E \Longrightarrow v = \sqrt{\left(\frac{2(1.106 \times 10^{-13})}{(4.00260)(1.66 \times 10^{-27})}\right)}$ 1

$v = 5.77 \times 10^6 \text{ m s}^{-1}$ 1

2

Total for question **7**

Explanation: This simple estimate assumes all the kinetic energy of the helium nucleus is available for the reaction.

Section C [30 marks]

18. (a) Describe the propagation of sound waves in air. [2]

Answer

Sound is a longitudinal wave: air molecules vibrate about their mean positions parallel to the direction of propagation 1

forming regions of compression and rarefaction that travel through the air (energy is transferred, not the air itself) 1

2

Explanation: Sound needs a medium; it cannot travel through a vacuum.

(b) A string of length L with mass per unit length, M, is stretched between two fixed points with a tension, T.

(i) Derive an expression for the frequency of the string when the string vibrates at the fundamental mode. [4]

Answer

Wave speed on the string: $v = \sqrt{\left(\frac{T}{M}\right)}$ 1

Fundamental: nodes at both ends, one loop: $L = \frac{\lambda}{2} \Longrightarrow \lambda = 2L$ 1

$f = \frac{v}{\lambda}$ 1

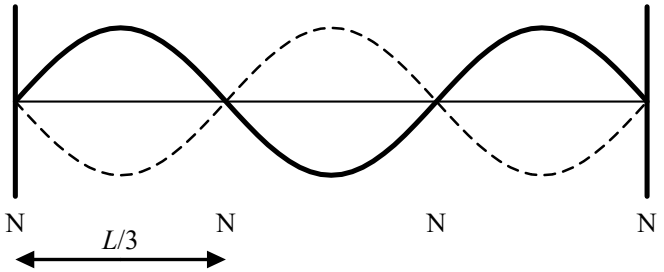
$f = \frac{1}{2L}\sqrt{\left(\frac{T}{M}\right)}$ 1

4

Explanation: This is Mersenne's law for the fundamental frequency of a stretched string.

(ii) Sketch the shape of the string for the second overtone and determine the distance between two consecutive nodes, in terms of the fundamental wavelength. [4]

Answer



Second overtone: three loops with nodes at both ends	2
Node spacing = $\frac{L}{3}$, and fundamental wavelength $\lambda_0 = 2L$	1
Distance = $\frac{\lambda_0}{6} = 0.167\lambda_0$	1
	4

Explanation: The second overtone is the third harmonic, $3f_0$.

(c) An ambulance moves at speed 42 m s^{-1} and sounds the siren at the frequency of 960 Hz. A bus which is in front of the ambulance stops at the roadside upon hearing the siren.
[Speed of sound in air is 330 m s^{-1}.]

(i) Determine the frequency of the siren heard by the passengers in the bus. [2]

Answer

Source approaching a stationary observer: $f' = f\frac{v}{v - v_s} = 960\frac{330}{330 - 42}$	1
f' = 1100 Hz	1
	2

Explanation: The ambulance is behind the bus and moving towards it, so the frequency heard is higher.

(ii) Determine the wavelength of the sound waves reflected by the bus and heard by the ambulance driver. [2]

Answer

The bus reflects 1100 Hz as a stationary source; the driver moves towards it: $f'' = 1100 \frac{330 + 42}{330} = 1240$ Hz — 1

$\lambda = \frac{v}{f''} = \frac{330}{1240} = 0.266$ m — 1

[The wavelength of the reflected sound in the air itself is 330/1100 = 0.300 m; 0.266 m is the wavelength corresponding to the frequency the driver hears.]

2

Explanation: The reflected sound undergoes a second Doppler shift because the driver (observer) moves towards the reflector.

(iii) State the change in the frequency heard by the passengers of the bus after the ambulance overtakes the bus. [1]

Answer

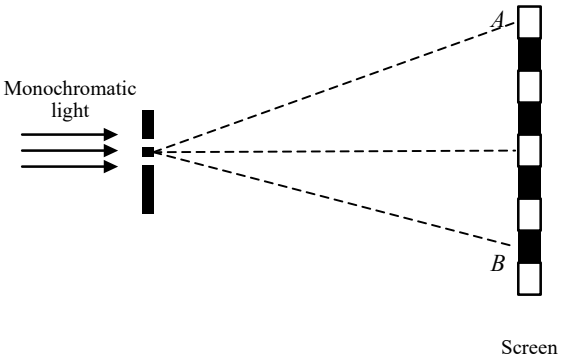
The frequency heard drops suddenly below 960 Hz (the ambulance is now moving away) — 1

1

Total for question **15**

Explanation: Moving away: $f' = 960 \times 330/(330 + 42) = 852$ Hz.

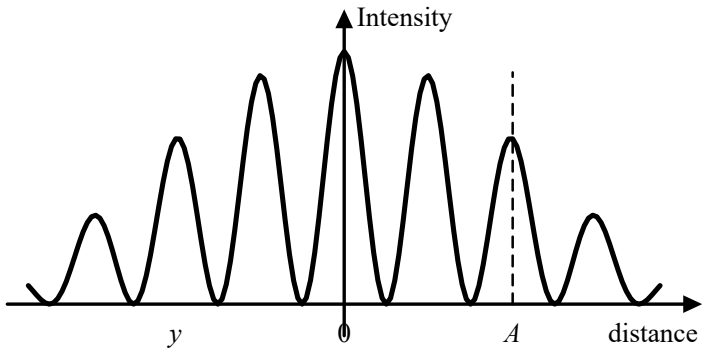
19. (a) A parallel beam of monochromatic light is passed through two slits as shown in the diagram below.



Both slits are separated by the distance of 1.4×10^{-4} m and the distance of the slits from the screen is 2.5 m. The bright fringe at A is 2.07×10^{-2} m from the central fringe.

(i) Sketch a graph to show the variation of light intensity with distance from the central fringe on the screen. [2]

Answer



Equally spaced peaks (bright fringes) separated by zero intensity (dark fringes)	1
Bright fringes of (nearly) equal intensity, maximum at the centre	1
	2

Explanation: For two narrow slits the intensity varies as $\cos^2$, slightly reduced away from the centre by single-slit diffraction.

(ii) Determine the frequency of the light which passes through the slits. [4]

Answer

A is the second bright fringe from the centre: $y_A = 2\frac{\lambda D}{d}$	1
$\lambda = \frac{(2.07 \times 10^{-2})(1.4 \times 10^{-4})}{2(2.5)} = 5.80 \times 10^{-7}$ m	1
$f = \frac{c}{\lambda} = \frac{3.00 \times 10^{8}}{5.80 \times 10^{-7}}$	1
$f = 5.18 \times 10^{14}$ Hz	1
	4

Explanation: Count the bright fringes on the diagram: A is two fringe separations from the central fringe.

(iii) Determine the path difference between the light that arrives at B from the two slits. [3]

Answer

B is the second dark fringe from the centre — 1

Path difference = $(n + ½)\lambda = 1.5\lambda$ — 1

$= 1.5(5.80 \times 10^{-7}) = 8.69 \times 10^{-7}$ m — 1

3

Explanation: Dark fringes occur where the waves arrive in antiphase: path difference ½λ, 1½λ, …

(b) Monochromatic light with wavelength 572 nm is incident from air on a thin soap film. The thin soap film has a refractive index of 1.30.

(i) Calculate the wavelength of the light within the thin soap film. [3]

Answer

The frequency is unchanged, but the speed in the film is c/n — 1

$\lambda_{film} = \frac{\lambda}{n} = \frac{572}{1.30}$ — 1

$\lambda_{film} = 440$ nm — 1

3

Explanation: Light slows down in a denser medium, so its wavelength shortens.

(ii) Calculate the minimum thickness of the thin soap film that produces a constructive interference for the reflected light.
[The refractive index of air is 1.00.] [3]

Answer

Reflection at the top surface (air → soap) has a phase change of π; at the bottom surface (soap → air) none — 1

Constructive: $2nt = (m + ½)\lambda$; minimum $t = \frac{\lambda}{4n}$ — 1

$t = \frac{572}{4(1.30)} = 110$ nm — 1

3

Total for question **15**

Explanation: Equivalently, $2t = ½\lambda_{film}$, so $t = \lambda_{film}/4 = 110$ nm.

20. X-rays are produced by accelerating electrons through a potential difference of 30.0 kV towards a metal target.

(a) Explain the production of the characteristic line spectrum and continuous spectrum in X-rays. [5]

Answer

Continuous spectrum: electrons hitting the target are decelerated by the nuclei of target atoms	1
They lose different amounts of kinetic energy, emitted as X-ray photons with a continuous range of wavelengths (bremsstrahlung)	1
The shortest wavelength corresponds to an electron losing all its kinetic energy in one collision	1
Line spectrum: incident electrons knock out electrons from inner shells (e.g. K shell) of target atoms	1
Electrons from higher shells fall into the vacancies, emitting photons of definite energies characteristic of the target (K_α, K_β)	1
	5

Explanation: The continuous spectrum depends on the accelerating voltage; the characteristic lines depend on the target element.

(b) Determine the shortest X-ray wavelength emitted. [4]

Answer

$eV = \frac{hc}{\lambda_{min}}$	1
$\lambda_{min} = \frac{hc}{eV}$	1
$= \frac{(6.63 \times 10^{-34})(3.00 \times 10^{8})}{(1.60 \times 10^{-19})(30.0 \times 10^{3})}$	1
$= 4.14 \times 10^{-11}$ m	1
	4

Explanation: All of the electron's kinetic energy, eV, becomes one photon.

(c) By using the wavelength in (b), the X-ray beam produces a first order Bragg maximum at an angle of 7.0° with the crystal planes. Determine the spacing of the crystal planes. [2]

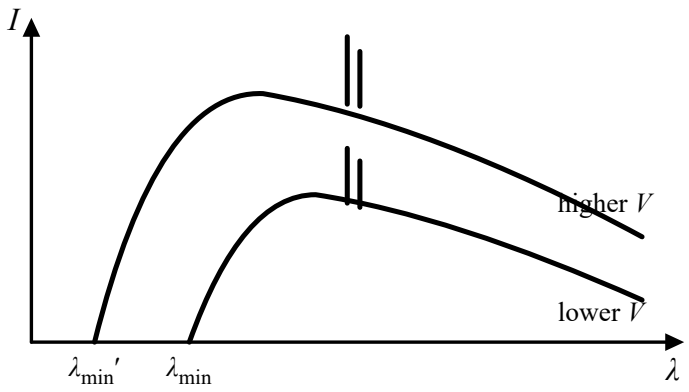
Answer

Bragg's law: $2d \sin\theta = n\lambda \Longrightarrow d = \frac{4.14 \times 10^{-11}}{2 \sin 7.0°}$	1
$d = 1.70 \times 10^{-10}$ m	1
	2

Explanation: The angle in Bragg's law is measured from the crystal planes (the glancing angle).

(d) With the aid of a graph, compare the X-ray spectra if a higher potential difference is used. [4]

Answer



Graph: both spectra drawn and labelled	1
Higher V: smaller minimum wavelength (cut-off shifts to the left)	1
Higher V: greater intensity at all wavelengths, and the peak of the continuous spectrum shifts to a shorter wavelength	1
The characteristic lines remain at the same wavelengths (same target)	1
	4
Total for question	**15**

Explanation: $\lambda_{min} = hc/eV$ is inversely proportional to the voltage; the line wavelengths depend only on the target metal.